THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Full Syllabus (Chapters 1–11) | Class 7 Science | Maximum Marks: 50

MULTIPLE CHOICE QUESTIONS — Answers 50 × 1 = 50

Q.	Correct Option	Brief Reasoning
1	**(d)** acidic	Sour taste is a property of acids.
2	**(b)** an electric cell	A cell has chemicals that produce electric current.
3	**(b)** lustre	Metals reflect light.
4	**(b)** physical change	Only form or state changes.
5	**(c)** puberty	Puberty marks the beginning of reproductive maturity.
6	**(c)** lower temperature	Heat flows downhill in temperature.
7	**(b)** second	1 minute = 60 seconds.
8	**(b)** ingestion	Ingestion is intake.
9	**(c)** autotrophs	Auto means self.
10	**(c)** straight lines	Rectilinear propagation.
11	**(c)** west to east	Counter-clockwise from above the North Pole.
12	**(b)** sulphuric acid	Sulphuric acid is a strong mineral acid.
13	**(c)** conductors	E.g. copper.
14	**(d)** brittle	They break when hammered.
15	**(d)** melting of wax	Wax solidifies again.
16	**(d)** puberty	Reproductive maturity.
17	**(b)** pass on heat to neighbours but stay in position	They vibrate.
18	**(c)** distance travelled per unit time	Speed = distance/time.
19	**(d)** villi	They increase surface area.
20	**(c)** blue-black	Starch test.
21	**(c)** a source of light, an opaque object and a screen	All three.
22	**(d)** 21 June	Summer solstice.
23	**(a)** base	Bases turn red litmus blue.
24	**(d)** hot	Heating effect of current.
25	**(d)** metal oxide	E.g. magnesium oxide.
26	**(c)** air and moisture	Both oxygen and water.
27	**(a)** growth spurt	Sudden increase.
28	**(a)** more	They absorb more radiation.
29	**(d)** odometer	Shows kilometres covered.
30	**(a)** breathing	Intake of O2 and release of CO2.
31	**(b)** leguminous plants	They fix nitrogen.
32	**(b)** reflection	Reflection.
33	**(c)** 27 days	Orbital period.
34	**(a)** basic	Bases are bitter and slippery.
35	**(d)** battery	A battery is a combination of cells.
36	**(a)** malleability	E.g. aluminium foil.
37	**(c)** chemical change	New substances appear.
38	**(b)** 11 to 19 years	Teenage years.
39	**(d)** conduction, convection, radiation	Standard modes.
40	**(d)** 3600 seconds	60 x 60.

41	**(a)** gills	Dissolved oxygen.
42	**(d)** make food for all	Base of the food chain.
43	**(b)** virtual and erect	Cannot be on a screen.
44	**(d)** Moon	Moon blocks Sun.
45	**(d)** H_2SO_4	It contains two hydrogen atoms.
46	**(d)** insulators	E.g. rubber.
47	**(a)** oxygen	Oxygen is a gas.
48	**(c)** ripening of fruit	Cannot be undone.
49	**(b)** voice box (larynx)	Larynx enlarges.
50	**(d)** moving particles of the fluid	Moving fluid.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Full Syllabus (Chapters 1–11) | Class 7 Science | Maximum Marks: 50

MULTIPLE CHOICE QUESTIONS — Answers — 50 × 1 = 50

Q.	Correct Option	Brief Reasoning
1	**(b)** metre per second	m/s.
2	**(b)** digestion	Digestion.
3	**(b)** photosynthesis	Photo means light.
4	**(b)** luminous objects	E.g. the Sun.
5	**(d)** 24 hours	It causes day and night.
6	**(b)** do not change the colour of litmus	They are neither acidic nor basic.
7	**(d)** electric iron	Heating element.
8	**(c)** basic	They turn red litmus blue in water.
9	**(b)** hydrated iron oxide	Reddish-brown flakes.
10	**(a)** girls	Girls begin earlier.
11	**(c)** light-coloured clothes	They reflect heat.
12	**(d)** speedometer	Shows km/h.
13	**(a)** liver	It makes bile.
14	**(b)** xylem	Xylem.
15	**(a)** closer to the light source	It blocks more light.
16	**(b)** 22 December	Winter solstice.
17	**(a)** lemon juice	Lemon juice contains citric acid.
18	**(a)** electric circuit	A complete path is needed.
19	**(d)** ductility	E.g. copper wire.
20	**(c)** physical change	Same substance, different state.
21	**(c)** teenage	Ages 13 to 19 are teen years.
22	**(d)** conduction	Heat passes from particle to particle.
23	**(d)** time period	Time for one to-and-fro swing.
24	**(a)** nostrils	Nose.
25	**(d)** fixing atmospheric nitrogen	Provide nitrates.
26	**(c)** a mirror	Polished surface.
27	**(b)** 29.5 days	Lunar month.
28	**(a)** $NaOH$	A common strong base.
29	**(d)** copper	Metals conduct.
30	**(b)** bromine	Bromine is a reddish-brown liquid.
31	**(c)** burning of magnesium ribbon	New substance forms.
32	**(d)** Adam's apple	A lump in the throat.
33	**(b)** radiation	Waves can travel in vacuum.
34	**(a)** kilometre per hour	km/h.
35	**(d)** moist skin	Skin.
36	**(c)** plants	Plants make food.
37	**(d)** same as the object	Equal.
38	**(d)** new moon day	Aligned.
39	**(a)** base	Turmeric becomes red with soap solution.
40	**(d)** nichrome wire	High resistance wire that gets very hot.
41	**(d)** acidic	E.g. sulphur dioxide.
42	**(d)** painting	Coating blocks air and water.

43	**(b)** bathe and keep clean	Hygiene.
44	**(a)** woollen clothes	Wool traps air.
45	**(a)** distance-time graph	Plots distance vs time.
46	**(a)** absorption	Absorption.
47	**(b)** chlorophyll	It traps sunlight.
48	**(b)** the Sun	It produces light.
49	**(a)** 365 and a quarter days	One year.
50	**(a)** baking soda solution	Baking soda is a base.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Full Syllabus (Chapters 1–11) | Class 7 Science | Maximum Marks: 50

MULTIPLE CHOICE QUESTIONS — Answers 50 × 1 = 50

Q.	Correct Option	Brief Reasoning
1	**(c)** a closed circuit	The bulb glows.
2	**(b)** sonority	E.g. bells.
3	**(b)** chemical change	Ash and gases form.
4	**(c)** hormones	Hormones travel in blood.
5	**(a)** convection	Hot fluid rises and cold falls.
6	**(c)** oscillation	Bob returns to the starting point.
7	**(d)** gall bladder	It helps digest fats.
8	**(b)** phloem	Phloem.
9	**(b)** umbra	Complete shadow.
10	**(c)** 21 March and 23 September	Equinoxes.
11	**(c)** $\mathrm{Ca(OH)_2}$	It is a base.
12	**(a)** plastic	Plastic does not conduct.
13	**(a)** iodine	Iodine has a shiny appearance.
14	**(c)** stretching a rubber band	Shape only.
15	**(c)** secondary sexual character	It appears at puberty.
16	**(b)** convection currents	Hot water rises.
17	**(a)** distance / time	Speed.
18	**(b)** lungs	They exchange gases.
19	**(d)** symbiosis	Both benefit.
20	**(b)** incident ray	Incident.
21	**(b)** new moon	Amavasya.
22	**(b)** acid	Acidic solutions turn it pink.
23	**(b)** fuse	It protects appliances.
24	**(a)** magnesium oxide	$2\,\mathrm{Mg} + \mathrm{O_2} \longrightarrow 2\,\mathrm{MgO}$.
25	**(a)** zinc	It protects iron.
26	**(a)** oil glands becoming more active	Hormones increase oil.
27	**(d)** traps air, a poor conductor	Trapped air insulates.
28	**(d)** straight line	Constant speed gives a straight line.
29	**(a)** heart	Heart.
30	**(c)** photosynthesis	Plants release.
31	**(d)** equal to	Equal behind the mirror.
32	**(c)** Earth	Earth's shadow falls on the Moon.
33	**(b)** sugar solution	Sugar solution has no acidic or basic character.
34	**(c)** an open circuit	No current flows.
35	**(a)** copper	Metals conduct.
36	**(c)** chemical change	Iron oxide forms.
37	**(c)** endocrine glands	Glands without ducts.
38	**(d)** radiation	E.g. from the Sun.
39	**(d)** length	Longer pendulum, longer period.
40	**(a)** egestion	Egestion.
41	**(c)** carbon dioxide and water	Plus sunlight.
42	**(b)** the Moon	It reflects sunlight.

43	**(b)** rotation of the Earth	Half of the Earth faces the Sun.
44	**(d)** antacid	It contains magnesium hydroxide.
45	**(a)** graphite	Graphite conducts.
46	**(b)** high melting points	Most metals melt at high temperatures.
47	**(c)** the moon's phases	Repeats at regular intervals.
48	**(a)** development of breasts	It appears at puberty.
49	**(a)** a metal rod in a flame	Metal rod.
50	**(c)** 60 km/h	120/2.

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